

## **Cyber Forensics & Security - Short Notes (5 Units)**

# Unit 1: Cyber Forensics Basics

- Cyber Forensics: Scientific process of identifying, collecting, analyzing and preserving digital evidence.
- Importance: Investigates cybercrime, supports legal cases, recovers data, prevents attacks.
- Scope: Computer, mobile, network, email, malware and cloud forensics.
- Types of Cyber Threats: Malware, phishing, ransomware, data breaches, insider threats.
- Digital Evidence: Emails, logs, browser history, files, call records, social media data.

## Important Exam Topics:

- Q. Explain Cyber Forensics.
- Q. Explain Importance.
- Q. Explain Scope.

## Unit 2: Vulnerabilities, Threats & Defense

- Vulnerability: Weakness exploitable by attackers.  $\text{Risk} = \text{Threat} + \text{Vulnerability} + \text{Impact}$ .
- Categories: Software, Hardware, Configuration vulnerabilities.
- Lifecycle: Discovery → Assessment → Prioritization → Remediation → Verification → Monitoring.
- Security Controls: Preventive, Detective, Corrective controls.
- Cyber Kill Chain: Reconnaissance to Actions on Objectives.

### Important Exam Topics:

- Q. Explain Vulnerability.
- Q. Explain Categories.
- Q. Explain Lifecycle.

## **Unit 3: IDS/IPS Security Systems**

- IDS monitors and alerts on suspicious behavior.
- Types: NIDS, HIDS, Hybrid IDS.
- Detection: Signature-based, Anomaly-based, Behavioral detection.
- IPS detects and actively blocks malicious traffic in real time.
- IPS Layers: Perimeter, Internal Segmentation, Application Defense.

### **Important Exam Topics:**

- Q. Explain IDS monitors and alerts on suspicious behavior..
- Q. Explain Types.
- Q. Explain Detection.

## **Unit 4: Web Automation, Scraping & Information Structuring**

- Web-based Automated Systems perform tasks automatically through internet interfaces.
- Core Components: Web Interface, Automation Engine, Data Storage, Integration Layer.
- Web Scraping Process: Request, Parse, Transform, Store.
- Information Structures: Hierarchical, Sequential, Matrix.
- E-commerce order processing as automation example.

### **Important Exam Topics:**

Q. Explain Web-based Automated Systems perform tasks automatically through internet interfaces..

Q. Explain Core Components.

Q. Explain Web Scraping Process.

## **Unit 5: Security Policies & Strategic Concepts**

- Strategic Decision Making: Long-term high-impact organizational decisions.
- Security Policies define rules; Procedures define implementation steps.
- Policy Components: Purpose, Roles, Compliance, Enforcement.
- Access Control Procedure: Requirement, Levels, Authentication, Documentation, Implementation.
- Confidentiality uses encryption, access controls and data classification.

### **Important Exam Topics:**

Q. Explain Strategic Decision Making.

Q. Explain Security Policies define rules; Procedures define implementation steps..

Q. Explain Policy Components.

# Quick Revision Formula Sheet

- Risk = Threat + Vulnerability + Impact
- Vulnerability Lifecycle: Discover → Assess → Fix → Monitor
- Cyber Kill Chain 7 Steps
- CIA Triad: Confidentiality, Integrity, Availability
- IDS = Detect | IPS = Detect + Prevent